

Correlation Analysis and Statistical Significance

A correlation matrix was generated to identify relationships among numerical variables and booking cancellations.

Key Findings

- Lead Time shows a positive correlation with cancellations, suggesting that advance bookings are more likely to be canceled.
- Booking Changes exhibit a negative correlation with cancellations, indicating that guests who modify reservations are generally more committed to completing their stay.
- Repeated Guests demonstrate lower cancellation tendencies compared to first-time customers.
- ADR exhibits only a weak correlation with cancellations, suggesting that pricing alone is not a strong predictor of cancellation behavior.

Statistical Significance

Correlation values indicate the strength and direction of relationships between variables; however, statistical significance determines whether these relationships are likely to be genuine rather than occurring by chance.

For categorical variables, Chi-Square testing was performed to validate observed relationships. Variables with p-values below 0.05 were considered statistically significant predictors of booking cancellations.

Conclusion

The combined correlation and statistical testing results suggest that Lead Time, Market Segment, Customer Type, and Booking Changes are among the most influential factors affecting hotel booking cancellations. These variables should be prioritized in predictive modeling and revenue optimization strategies.